1st Semester 2024 TST

Paly Math Club

October 2, 2024

This test is the team-selection test for **Berkeley Math Tournament**, **Harvard-MIT Math Tournament**, and **Caltech Math Meet**. The following are brief informations of both competitions:

BMT	HMMT	CMM
Date: Nov 2nd, 2024	Date: Nov 9th, 2024	Jan 25, 2025
Cost: $0	Cost: $250 - $350	Cost: $150 - $250
Location: UC Berkeley, Berkeley CA	Location: Harvard, Boston, MA	Location: Caltech, Los Angeles CA

Please circle the competitions you are planning to attend (This decision is **binding**):

1) BMT 2) HMMT 3) CMM

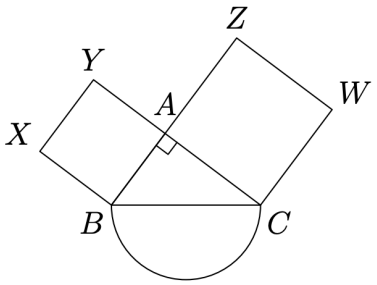
Instructions: This is a **60** minute test with **15** questions. Each question is given a number of points, which reflects the intended difficulty of each problem, and you will be ranked based on your total number of points. No point deduction for incorrect answers, no partial credit.

Name: _____

1. How many digits does the product $2^{42} * 5^{38}$ have? (1pt)

2. A *festive* number is a four digit integer containing one of each of the digits 0, 1, 2, and 4 in its decimal representation. How many *festive* numbers are there? (1pt)

3. Right triangle ABC has squares $ABXY$ and $ACWZ$ drawn externally to its legs and a semicircle drawn externally to its hypotenuse BC. If the area of the semicircle is 18π and the area of triangle ABC is 30, what is the sum of the areas of squares $ABXY$ and $ACWZ$? (1pt)



4. Find the smallest number with exactly 28 divisors. (2pt)

5. Billy the kid likes to play on escalators that move up in constant speed. Going at a constant speed, he manages to climb up one escalator in 24 seconds and climb back down the same escalator in 40 seconds. If at any given moment the escalator contains 48 steps, how many steps can Billy climb in one second? (2pt)

6. A ball inside a rectangular container of width 7 and height 12 is launched from the lower-left vertex of the container. It first strikes the right side of the container after traveling a distance of $\sqrt{53}$ (and strikes no other sides between its launch and its impact with the right side). How many times does the ball bounce before it returns to a vertex? (The final contact with a vertex does not count as a bounce) (2pts)

7. Cyclic quadrilateral ABCD has side lengths $AB = 6, BC = 7, CD = 7, DA = 6$. What is the area of $ABCD$? (3pts)

8. Let a and b be the roots of the polynomial $x^2 + 2020x + c$. Given that $\frac{a}{b} + \frac{b}{a} = 98$, compute $\sqrt{c}$ (3pts).

9. In ΔABC, $AB = 2019, BC = 2020,$ and $AC = 2021$. Sky draws three regular n-gons in the plane of ΔABC so that each n-gon shares a side with a distinct side of ΔABC and no two of the n-gons overlap. What is the maximum possible value of n? (3pts)

10. Let real numbers a, b, c, d satisfy the following system of equations:

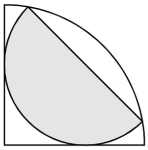
$$\begin{aligned}\log_2 a + \log_3 b + \log_4 c &= 13\\ \log_2 a + \log_3 b + \log_5 d &= 11\\ \log_2 a + \log_4 c + \log_5 d &= 15\\ \log_3 b + \log_4 c + \log_5 d &= 12\end{aligned}$$

Find the value of $a+b+c+d$. (4pts)

11. Find the value of
$$\sum_{a=1}^{\infty}\sum_{b=1}^{\infty}\sum_{c=1}^{\infty}\frac{ab(3a+c)}{4^{a+b+c}(a+b)(b+c)(c+a)}$$
(4pts)

12. Find the area of the set of all points z in the complex plane that satisfy
$$|z-3i|+|z-4|\leq 5\sqrt{2}.$$
(4pts)

13. A semicircle is inscribed in a quarter circle as shown. What is the fraction of the quarter circle shaded?



(4pts)

14. Let sequence $\{a_i\}_{i=0}^{\infty}$ be defined by $a_0=\frac{1}{2}$ and $a_n = 1+(a_{n-1}-1)^2$. Find the product
$$\prod_{i=0}^{\infty} a_i = a_0a_1a_2\ldots$$
(5pts)

15. The equation
$$x^{10}+(13x-1)^{10}=0$$
has 10 complex roots $r_1, \bar{r_1}, r_2, \bar{r_2}, r_3, \bar{r_3}, r_4, \bar{r_4}, r_5, \bar{r_5}$, where the bar denotes complex conjugation. Find the value of
$$\frac{1}{r_1\bar{r_1}}+\frac{1}{r_2\bar{r_2}}+\frac{1}{r_3\bar{r_3}}+\frac{1}{r_4\bar{r_4}}+\frac{1}{r_5\bar{r_5}}.$$
(5pts)